

1. Online Hitting Set for Intervals

(a) **Sophia:** $\Omega(n)$

$S = \{1, \dots, n\}$. Adversary:

$$A_i = [1, n-i+1], \quad i = 1, \dots, n$$

Sophia: $\forall i: A_i \cap H_{i-1} = \emptyset \implies \text{add } \max(S \cap A_i) = n-i+1$.

$$H_{\text{Sophia}} = \{1, 2, \dots, n\}, \quad |H_{\text{Sophia}}| = n$$

OPT: $1 \in A_i \forall i \implies H^* = \{1\}, |H^*| = 1$.

$$\boxed{\text{ALG/OPT} = n = \Omega(n)}$$

(b) **Tammy:** $O(\log n)$

Tammy adds $m = \lceil (\min(S \cap A) + \max(S \cap A))/2 \rceil$ when $A \cap H = \emptyset$.

Let $H^* = \{p_1^*, \dots, p_k^*\}$ be OPT. $\forall A_i$ with $A_i \cap H_{i-1} = \emptyset: \exists p^* \in H^* \cap A_i$.

Window: $\forall p^* \in H^*$, define $W_{p^*} \subseteq S$ = points strictly between nearest added points flanking p^* . Initially $W_{p^*} = S$.

Halving: fix $p^* \in H^* \cap A_i$. $A_i \cap H_{i-1} = \emptyset \implies A_i \cap S \subseteq W_{p^*}$. m midpoint of $A_i \cap S$, $p^* \in A_i \implies$

$$|W_{p^*}^{\text{new}}| \leq \frac{1}{2}|A_i \cap S| \leq \frac{1}{2}|W_{p^*}^{\text{old}}|$$

$|W_{p^*}| \leq n$ initially $\implies \leq \lceil \log_2 n \rceil$ such A_i per p^* .

$$|H_{\text{Tammy}}| \leq \sum_{p^* \in H^*} \lceil \log_2 n \rceil = k \lceil \log_2 n \rceil$$

$$\boxed{\text{ALG/OPT} \leq \lceil \log_2 n \rceil = O(\log n)}$$

2. Online Unit Covering

(i) **Leftmost:** unbounded

Adversary: $\varepsilon \in (0, 1/n)$, $x_i = -(i-1)\varepsilon$.

$\forall j < i: x_j = -(j-1)\varepsilon > -(i-1)\varepsilon = x_i \implies x_i < x_j \implies x_i \notin [x_j, x_{j+1}]$. So Leftmost adds $[x_i, x_{i+1}]$ for every i .

$$|\text{ALG}| = n$$

OPT: all $x_i \in [-(n-1)\varepsilon, 0] \subseteq [-(n-1)\varepsilon, 1-(n-1)\varepsilon]$ (length 1 since $(n-1)\varepsilon < 1$). $|\text{OPT}| = 1$.

$$\boxed{\text{ALG/OPT} = n \rightarrow \infty}$$

(ii) Centered: 2-competitive

Let $C = (c_1, \dots, c_a)$ = centers in arrival order.

Claim 1: $\forall j < k: |c_k - c_j| > 1/2$.

$$c_k \text{ uncovered} \wedge [c_j - \frac{1}{2}, c_j + \frac{1}{2}] \in \mathcal{C} \implies c_k \notin [c_j - \frac{1}{2}, c_j + \frac{1}{2}] \implies |c_k - c_j| > \frac{1}{2}$$

Claim 2: $\forall I$ unit interval: $|C \cap I| \leq 2$.

$$c_p < c_q < c_r \in I \implies c_r - c_p > 1 \quad (\text{Claim 1: } c_q - c_p > \frac{1}{2}, c_r - c_q > \frac{1}{2}) \quad \text{but } c_r - c_p \leq 1$$

$\forall c \in C : \exists I^* \in \text{OPT}$ with $c \in I^*$ (OPT covers c). Each I^* contains ≤ 2 centers.

$$\boxed{\text{ALG} = |C| \leq 2 \cdot |\text{OPT}|}$$

(iii) Grid: 2-competitive

Cell $k = [k, k+1)$, $k \in \mathbb{Z}$.

Claim: $\forall I^* = [a, a+1] \in \text{OPT}$: $|\{k : I^* \cap \text{cell } k \neq \emptyset\}| \leq 2$.

$$|I^*| = 1 \implies I^* \subseteq \text{cell } \lfloor a \rfloor \cup \text{cell } (\lfloor a \rfloor + 1)$$

ALG opens ≤ 1 interval per cell. $\forall x \in \text{points} : \exists I^* \ni x \implies$ each ALG cell mapped to some OPT interval, each OPT interval $\rightarrow \leq 2$ cells.

$$\boxed{\text{ALG} \leq 2 \cdot \text{OPT}}$$

3. Online Unit Clustering

Algorithm (Grid)

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begin
  GRIDCLUSTER( $x_1, x_2, \dots$ )
   $\Pi \leftarrow \emptyset$  // partition into clusters, indexed by integer label
  for  $i \leftarrow 1, 2, \dots$  do
     $k \leftarrow \lfloor x_i \rfloor$ 
    if  $\exists C_k \in \Pi$  with label  $k$  then
       $C_k \leftarrow C_k \cup \{x_i\}$ 
    else
       $\Pi \leftarrow \Pi \cup \{(\{x_i\}, k)\}$  // new cluster, label  $k$ 
  return  $\Pi$ 

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Validity: cluster labeled k has all points $\in [k, k+1) \implies$ diameter $< 1 \implies$ coverable by $[k, k+1]$.

Competitive analysis

OPT: k^* clusters $C_1^*, \dots, C_{k^*}^*$. $\forall j: \exists a_j \in \mathbb{R}$ with $C_j^* \subseteq [a_j, a_j+1]$.

Claim: $\forall j: |\{ \lfloor x \rfloor : x \in C_j^* \}| \leq 2$.

$$x \in C_j^* \subseteq [a_j, a_j+1] \implies \lfloor x \rfloor \in \{ \lfloor a_j \rfloor, \lfloor a_j \rfloor + 1 \}$$

$|\text{ALG}| = |\{ \lfloor x_i \rfloor \}_i|$. Each OPT cluster $\rightarrow \leq 2$ ALG labels:

$$|\text{ALG}| \leq \sum_{j=1}^{k^*} 2 = 2k^*$$

$$\boxed{\text{ALG/OPT} \leq 2}$$